
Melioidosis: A Global Health Threat and Innovative Strategies for Treatment and Prevention

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Untreated, this disease can rapidly advance and the person is at a high risk of dying. Acute cases have a death rate of 40–50%; *Burkholderia pseudomallei* resists many commonly used antibiotics, thus raising many treatment complications.

Abstract

Burkholderia pseudomallei is the cause of the dangerous infection melioidosis. A bacterium known *Burkholderia pseudomallei* lives tropical and subtropical soils and waters. Both humans and animals typically contract the disease by skin wounds, contaminated dust inhalation, or contaminated drinking water use. Melioidosis occurs most frequently during flooding or heavy rains. Stagnant water allows the pathogen to invade. If left untreated, the patient can die quickly, as the disease can move fast. Approximately 40% to 50% of cases report death, especially in the acute forms of the disease.

The typically infectious route is inhalation, through wounds or via ingestion with risk factors, such as diabetes, renal disease or respiratory disease. The presentations are extremely diverse with possible symptoms of mild fever, pneumonia and death through septic shock and multi-organ failure. Diagnosis does not lend itself to easy diagnosis and requires sophisticated laboratory process. Moreover, these reports implore us to think about volumes of complicated, even resistant infections, to antibiotics traditionally used. In total, lengthy (~6-12 week) courses with different antibiotics are often required.

Alongside classical therapy, there are numerous complementary treatments under consideration. Garlic (Allicin) has some antibacterial activities due to its interference with the lipopolysaccharides and the bacterial membrane and, hence, can very well be a supportive therapy with very minor side effects. Syringaldehyde (plant phenolic aldehyde) inhibits the Type III secretion system, thus disturbing the bacterial membrane and creating oxidative stress, hence may be useful as an adjunct.

Because it is a global health concern, melioidosis continues to present a challenge and must have new approaches to therapy, prevention, and international collaborative efforts to lessen its weight.

Keywords:

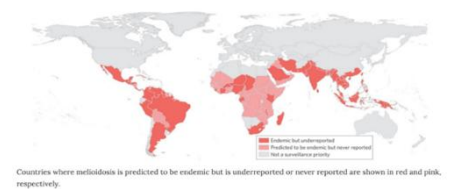
Melioidosis; *Burkholderia pseudomallei*; Clinical manifestations; Antibiotic resistance; Garlic; Syringaldehyde; Prevention; Vaccine development

Introduction:

Melioidosis is a serious infectious disease that transforms from a common infectious disease from the infection called *Burkholderia pseudomallei*. This pathogen is found in soil and water, especially in tropical places like Southeast Asia and Northern Australia. This map may be useful for determining where to prioritize health surveillance activities and implement prevention strategies in regionally dependent manner.

Figure 3: Priority countries where microbiological diagnostic facilities and disease reporting systems for melioidosis should be strengthened.

From: Predicted global distribution of *Burkholderia pseudomallei* and burden of melioidosis



Both humans and animals usually get this disease through a skin cut, breathing in contaminated dust, drinking contaminated water, or two water droplets. This particular disease is contracted more during flood and heavy rain, where the soil and water has stagnant water infected with the pathogen. Untreated, this disease can rapidly advance and the person is at a high risk of dying. Acute cases have a death rate of 40–50%, unsupported, this rate can considerably expand.

In addition, the economic impact of this disease is that the most productive working men suffer from this in the endemic regions. The workers of this region, such as loggers and farmers, because of wet soil, are at a high risk of coming into contact with this disease. Chronic diseases like diabetes can increase risk in such individuals, as will a reduced immune system.

Dr. Alfred Whitmore was the first to describe the disease in 1911 in Burma (now Myanmar), and it is therefore named "Whitmore's disease." Since then, improvements in medicine have increased our understanding of some of the other means of transmission, in addition to its pathology and treatment challenges. Attention has been drawn to *Burkholderia pseudomallei* (as show in figure:1) due to its resistance to multiple antibiotics and its possible use as a biological warfare agent, which increases the need to create new therapeutic and preventive options.



Distribution of Melioidosis

Environmental Factors

These areas provide environmental conditions which favor survival and rapid increase of bacteria. Hot and humid conditions; wet soils; and standing water bodies such as rice fields and swamps for example, are conducive to bacterium survival and the ability to multiply. A wet season with heavy rain with the influence of flooding activities on soil and water, allows the proliferation and dispersion of the bacterium in soil and water; therefore, humans are more likely to encounter the bacterium than any other circumstance.

Incidence Rates

Incidence rates are very different departmentally and geographically; there may also be variations of annual incidence rates across the different areas within a country. For example, the annual incidence rates in Negeri Sembilan, Malaysia are between less than 2 and over 20 cases annually per 100,000 population based on epidemiological surveillance and rural vs urban distinctions. Furthermore, incidence rates tend to be higher in rural and agricultural regions of Malaysia where the population is likely to be directly exposed to contaminated soil and water.

Mortality Rates

Studies show that mortality rates from melioidosis in these areas are between 9-17%, and even reach 50%, in untreated or complicated cases. The incidence of melioidosis is influenced by environmental factors. The greatest environmental factor is rainfall, however temperature and rainy season intensity factor heavily as well. Cases increased in number after considerable rain events, from flooding which allows the bacterium to be more widely disseminated.

Modes of Transmission

- **Skin contact:** the most common route, where the bacterium is able to enter through cuts or scratches in the skin when coming into contact with contaminated soil or water.
- **Inhalation:** occurs when dust or aerosols are inhaled that contain the bacterium. Often people have contracted the disease during a flood or storm.
- **Ingestion:** is a relatively rare route, this occurs when someone drinks food or water contaminated with the bacterium.

The disease is transmitted mainly from the environment to human, with person-to-person transmission being rare. Incubation Period.

The incubation period is generally between one to 21 days, with an average period of 9 days, although in very rare instances, the bacterium may remain dormant for some time, which can be weeks or months and even years, before symptoms develop.

Modern diagnostic methods for melioidosis include culture-based techniques, molecular diagnostics, and radiological imaging.

- Culture:

Melioidosis is diagnosed by isolating *B. pseudomallei* from blood, urine, sputum, skin lesions, cerebrospinal fluid, oropharyngeal swab, rectal swab, or abscesses, or by detecting an antibody response to the bacteria. Most commonly, the morphological characteristics of *B. pseudomallei* and its colony as cultured on blood agar plates is investigated. One of its distinctive features is the “safety pin” structure, which is due to bipolar gram staining. However, this process is very time-consuming and can take up to 7 days to yield results. In addition to this delay, the culture offers poor sensitivity.

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- Radiological Imaging: X-rays, CT scans, and MRI help visualize infection in different organs, especially in pulmonary and disseminated forms.
 - Fine Needle Aspiration (FNA): Valuable for detecting melioidosis in localized infections like abscesses.
 - Indirect Hemagglutination Assay: An indirect hemagglutination assay (IHA) is also widely used for the serodiagnosis of melioidosis. The assay is based on the detection of antibodies against whole bacteria of *B. pseudomallei*.

Also, a new diagnostic for melioidosis, called Rapid Melioidosis, takes only 15 minutes and can be administered in a point-of-care (POC) setting. The test can detect infection at a very early stage from just one drop of blood or serum. Instead of a lengthy lab test that requires the bacteria to be isolated from a clinical specimen and cultured, the new test yields rapid results. Rapid Melioidosis is an immunochromatography test that uses an antigen for hemolysin-coregulated protein (Hcp1) to detect antibodies against Hcp1 in the blood. Hcp1 is a protein secreted by the bacteria that plays a role in its intracellular functions. Studies have shown that this new melioidosis test kit is highly sensitive and specific. The test was developed by researchers at Mahidol University in Bangkok, Thailand and at the University of Washington in Seattle, Washington. The test is currently licensed and available for commercial use in Thailand under the name Active Melioidosis Detect. The test is currently licensed by the Thailand Food and Drug Administration and available for commercial use in Thailand and other South Asian countries including Cambodia, Malaysia, Vietnam, Singapore and Indonesia under the name Mahidol University Tropical Medicine Melioidosis Antibody Test (MUTM Melioidosis Antibody Test). The current antigen is recombinant protein which is more sensitive and specific than the initial kit and is linked to a second O-polysaccharide for added specificity.

Mahidol University has also launched a real-time polymerase-chain-reaction (PCR) test kit, called Melimo, for identifying *B. pseudomallei* based on a Type 3 Secretion System Gene marker-1 (TTS1). Combining Hcp1-ICT with type-3 secretion system 1 quantitative PCR (TTS1-qPCR) significantly enhances diagnostic performance. The research team is also examining the use of the plasma IL-1R2 protein as an effective biomarker for predicting sepsis-related mortality and determining treatment regimens.

The importance of understanding distribution

The understanding of geographical distribution, infection dynamics and environmental factors associated with melioidosis is essential to implement prevention and surveillance measures in humid rural agricultural areas where melioidosis is more common.

Disease Spread and Transmission

The organism is found in high-humidity conditions and infects humans through many routes, including skin contact (particularly through cuts or scratches to contaminated soil or water), inhalation (primarily if inhaled during dust storms, cyclones or floods), and less commonly ingestion (from contaminated water or food).

Clinical Picture: Varied Symptoms

- Sudden onset usually after the incubation period of 1 to 21 days.
- Symptoms of high fever, chills, myalgia, headache, and dyspnea are seen that could easily be misconstrued with sepsis.
- It could lead to acute pneumonia, which would cause distress in breathing and other issues.
- It could result in abscess formation in organs such as the liver, spleen, kidneys, skin, bones and joints.
- severe cases, it leads to septic shock and multiorgan failure rapidly, especially in immunocompromised persons or aged people.

Chronic

- Can go on for a number of months or even years and simulate other chronic lung diseases like tuberculosis with symptoms of cough for a long time, loss of weight, night sweats, and general weakness.
- It has a slower pace and is less severe than the acute form but causes localized destructiveness, particularly of the lungs and bones.

Complications

- Double or triple abscesses in internal organs.
- Sepsis and multiple organ failure.
- Chronic pneumonia with Similarity with Other Diseases Diagnosis gets complicated as melioidosis is regarded as "great imitator" given

the symptom overlap with other diseases such as:

- Pulmonary tuberculosis.
- Bacterial pneumonia.
- Organ abscesses from other causes.
- Autoimmune diseases are sometimes confused with melioidosis and even with inflammatory bowel disease.

Traditional Laboratory Tests

Bacterial Culture: The gold standard of melioidosis diagnosis consists of culturing *Burkholderia pseudomallei* from blood, sputum, or other bodily fluids, abscesses, or infected tissues. This process may take several days

since the bacteria must be handled in specialized laboratories as the organism is considered a biosafety hazard level.

Molecular Tests

PCR: To identify the DNA of *Burkholderia pseudomallei* rapidly and accurately from clinical samples, PCR is useful especially in speeding up diagnosis of acute cases compared with culture. **Antibody Test:** Rarely used in support of diagnosis, but are less reliable because of a high rate of false positives, especially in endemic areas.

Difficulties in Resource-Limited Settings

- In many endemic countries, they rely on the traditional culture methods which require technical expertise

Treatment for *Burkholderia pseudomallei* Infection Causing Melioidosis

The treatment of infections caused by *Burkholderia pseudomallei* and ensuing melioidosis requires precise and intensive antibiotic protocols, considering the complexity of the disease, treatment duration, high antibiotic resistance of the bacteria, and hence, the management of acute cases. Current Antibiotic Protocols and Treatment Duration

Present-day Antibiotic Regimens and Duration of Treatment

Immediately, treatment starts with an induction phase consisting of the administration of intravenous antibiotics that are potent against the organism, e.g., Ceftazidime or Carbapenems, for about 10-14 days. This phase is necessary to control the initial infection and to rid the bacteria from the bloodstream or tissues.

The induction phase is followed by the maintenance or eradication phase with oral antibiotics, i.e., Trimethoprim/Sulfamethoxazole (TMP-SMX), an extended duration of 3 to 6 months with the intention to completely eradicate the bacteria and prevent its relapse.

In situations where the patient is immunocompromised or in an advanced case situation, even longer protocols of treatment might be appropriate, or other treatments according to the patient's condition should be considered.

Resistance to Antibiotics

Burkholderia pseudomallei resists many commonly used antibiotics, thus raising many treatment complications.

Targeted antibiotic therapy usually has to be administered, which are often second- or third-line agents; continuous susceptibility testing and effective antibiotic stewardship practice must be considered so as not to worsen the existing problem of antibiotic resistance.

Antibiotic resistance is an ever-building issue in the world, increasing mortality rates and rendering management of treatment harder, hence requiring recent advances in pharmaceuticals that could hopefully lead to finding an alternative.

Management of Critical Cases

Managing critical cases often means that there is the need for admission into the intensive care unit (ICU), especially in cases involving septic shock or failure of organs.

Supporting organ function comes near the top of extremely important things to do above or parallel to administering antibiotics as soon as possible.

Careful monitoring of inflammatory markers and clinical signs is also important.

Managing relapses and comprehensive care of the critically ill call for multidisciplinary medical teams.

Vaccine Development for Melioidosis: Progress and Research

No Licensed Vaccine Yet

As of now, no vaccine has been licensed for human use against melioidosis. This remains true both in endemic areas and for biodefense applications.

Live Attenuated Vaccines (LAVs)

LVS Δ capB-vectored vaccine: A live-attenuated vector based on the tularemia vaccine (LVS) modified with a Δ capB mutation. It expresses multiple *B. pseudomallei* antigens and has shown high efficacy in lethal respiratory challenge models in mice, with advantages like cost-effectiveness and ease of manufacturing.

B. thailandensis E555 Δ ilvI-based LAV: A genetically attenuated strain expressing similar capsular polysaccharides (CPS) to *B. pseudomallei*, showing protective immunity in mouse models.

Subunit & Conjugate Vaccines

Lipopolysaccharide (LPS)–Protein Conjugate: A conjugate vaccine combining *B. pseudomallei* LPS with the tetanus toxoid Hc fragment induced substantial protective immunity in mice, with strong humoral (IgG1 and IgG2a) and Th1-biased responses.

Subunit antigens: Proteins like CPS, LolC, Omp85, and Hcp2, as well as outer membrane vesicles, have emerged as promising subunit candidates in preclinical studies

In-Silico and Multi-Epitope Peptide Vaccines

A chimeric multi-epitope peptide vaccine was designed using immunoinformatics tools, showing broad theoretical population coverage (~99.7%) and predicted strong binding to human TLR-5, suggesting potential for robust immune response activation.

Next-Generation Platforms

Emerging vaccine platforms include:

- Adenoviral-vectored vaccines
 - mRNA vaccines
 - Peptide-based nanovaccines
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These methods offer advantages like low cost, scalability, strong humoral and cellular immunity, and adaptability in low-resource settings. However, they face challenges expressing non-protein antigens (e.g., polysaccharides) that are often key to protection.

6. Institutional Collaborations & Preclinical Advances

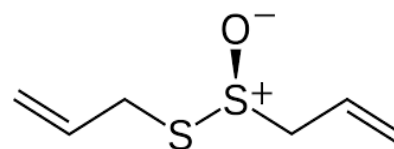
INRS (Canada): Developing a glycoconjugate (sugar-based) vaccine, linking bacterial polysaccharides to protein carriers, currently in preclinical (murine) testing.

University College Dublin (UCD): Developed a vaccine candidate dubbed MelioVac, licensed to Poolbeg Pharma for further development toward human trials; The DiSarM project identified a protective subunit vaccine in mice and aims to proceed through safety testing and diabetics-model evaluation toward clinical readiness.

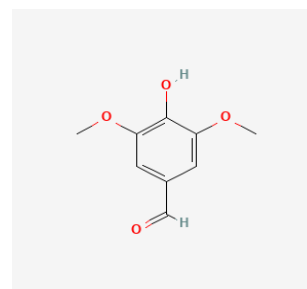
Our Solution:

Garlic & Syringaldehyde : Treatment and Benefits Without Side Effects

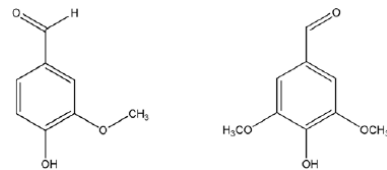
Allicin (as show in figure 2) a compound present in garlic, is said to possess antibacterial capabilities; in fact, it mainly acts against Gram-negative bacteria. Scientific studies support the premise that allicin targets the bacterial cell membrane by interfering with critical components, including lipopolysaccharides (LPS), thereby compromising bacterial survival and virulence. Despite its natural ability to combat infections, garlic or allicin shouldn't be used as a replacement for the approved antibiotic; instead, they could potentially reduce resistance or boost immune response with almost negligible side effects when compared to synthetic antibiotics. Recognition of garlic in therapy warrants much broader clinical trials, but as a complementary option[it] may hold some prospect for preventive[or] supportive therapy. Some advantages attributed to natural remedies, such as garlic, include their easy accessibility and low cost, combined with safety when used by humankind, while, of course, producing some benefit and immunological protection against a host of bacterial and viral afflictions.



As shown before in multiple studies the resistance of *Burkholderia pseudomallei* towards the antibiotics taken is real. Besides taking garlic which acts as the main antibiotic in the new treatment taking syringaldehyde, which is extracted from plants like *Manihot esculenta*, *Magnolia officinalis* and *Rhinacanthus*, would act as an active agent to suppress bacterial activities. Syringaldehyde (as show in figure 3) is a member of the phenolic aldehyde family. It helps in inhibiting type III secretion system (T3SS) which is a bacterial system used by the bacteria to secrete proteins into host cell cytoplasm to start colonizing host cells to escape phagosomes and extracellular barriers milieu. Syringaldehyde acts as an antibacterial and an antioxidant. This is because of the phenolic compounds in syringaldehyde so it can insert into bacterial cell membranes and disrupt integrity by disturbing lipid packing, leading to leakage of ions and



metabolites. Also the -CHO group can form Schiff bases with bacterial proteins (especially with lysine residues), altering enzyme function or inactivating metabolic enzymes. This covalent reactivity can directly affect bacterial metabolism. In addition, phenolic aldehydes organic compounds that feature both a hydroxyl group (-OH) attached to an aromatic ring (the "phenol" part) and an aldehyde group (-CHO) (as show in figure 4) sometimes generate reactive oxygen species (ROS) or interfere with redox homeostasis inside bacterial cells, creating oxidative stress that kills or inhibits the bacteria.



Prevention and Control

Personal preventive measures:

Keep away from contact with contaminated soil and water; especially tropical areas and infected plants are breeding grounds for *Burkholderia pseudomallei*.

When handling soil or water, always use personal protective equipment such as boots and gloves.

Maintain personal hygiene and wash your hands thoroughly with soap and running water.

Make sure that all wounds and scratches have a proper covering to minimize the chance of bacteria entering.

Consume garlic every day, as this will minimize the chance of any disease and help with immunity.

Garlic contains allicin that is naturally antibacterial and may help keep the body fight infections. Public Health Programs:

Conduct melioidosis monitoring in endemic areas and also improve case capture.

Increase the awareness and health education of the population, including health care workers on routes of transmission, prevention and treatment, but also the benefits of consuming garlic daily as a supportive measure that may help enhance immunity.

Investigate improving health care infrastructure that provides effective antibiotics and supports antibiotic resistance programs.

Awareness and Education:

Train health care teams to assess, diagnose and manage melioidosis.

Implement public awareness campaigns to inform why community members take precautions to prevent the spread of disease and improve health and also to increase the use of garlic daily as a natural remedy for the protection of their immunity.

Increase education for appropriate use of antibiotics to reduce the continue to spread of resistances.

Conclusion:

major public health issue globally, especially in tropical and subtropical countries where *Burkholderia pseudomallei* thrives in the natural environment. The disease presents a significant public health issue owing to its variable phenotypes, mortality of greater than %50, and diagnostic challenge, along with its variable course of estimated its unreliability as a public health target. Furthermore, the background resistance of *B. pseudomallei* to many commonly used antibiotics, and development of resistance to the growing number of antibiotics, is yet another impetus for revamping treatment options. Standard formal care involving prolonged intravenous and/or oral antibiotics will remain the key guidance to healthcare providers - inclusive of working with patients and families .

However, adjunct medicinal products, whereby an oregano, garlic (including Allicin), residual topically applied honey and syringaldehyde - with minimal adverse effects, antimicrobial and immunomodulating effects, and ability to inhibit bacterial secretion systems and induce oxidative stress, are a compelling adjunct, given that they do NOT replace antibiotics rather support antibiotic treatment and might even improve overall treatment outcome, reduce recurrence, and slow overall resistance development. In order to help reduce intermittent outbreaks in endemic areas, prevention from personal protective measures, awareness, and improved public health practices, are decisive. Secondly my recommendation: improve and accelerate international collaborative research response on diagnostics, vaccines and new therapeutics.

Resources:

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